Poisson Conference 2022

Program and abstracts

July, 25–29, 2022

ICMAT Madrid

Madrid, Spain

https://www.icmat.es/congresos/2022/poisson/index.php

Poisson 2022 is the 12th in a series of international conferences on Poisson geometry. The Poisson 2022 Conference will be held at the Headquarters of the Spanish National Research Council, located in Calle Serrano 117, 28006, Madrid, from July 25 to July 29, 2022. It is organized by the Institute of Mathematical Sciences in Madrid. The Poisson 2022 Advanced School will be held at the Centre de Recerca Matemàtica, Barcelona.

These activities are part of the Joint CRM-ICMAT Intensive Research Programme on Poisson Geometry, organised by the Institute of Mathematical Sciences in Madrid and the Centre de Recerca Matemàtica in Barcelona. For details on previous conferences, visit the Poisson Geometry Website.

The Andre Lichnerowicz Prize in Poisson geometry was established in 2008. It is awarded for notable contributions to Poisson geometry, every two years at the "International Conference on Poisson Geometry in Mathematics and Physics", to researchers who completed their doctorates at most eight years before the year of the Conference. The prize is named in memory of Andre Lichnerowicz (1915-1998) whose work was fundamental in establishing Poisson geometry as a branch of mathematics. It is awarded by a jury composed of the members of the scientific/advisory committee of the conference. The Lichnerowicz prizes 2022 will be sponsored by the University of Goettingen.

There will be a topical Issue on the Journal of Geometry and Physics dedicated to Poisson Geometry in Mathematics and Physics Editors: Anton Alekseev, Lisa Jeffrey, Eva Miranda and Volodya Roubtsov

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
VENUE ICMAT CAMPUS UAM 9:00 BUS DEPARTURE 9:30 REGISTRATION AT ICMAT	VENUE CSIC CENTRAL BUILDING	VENUE CSIC CENTRAL BUILDING	VENUE CSIC CENTRAL BUILDING	VENUE CSIC CENTRAL BUILDING
10:00 - 10,30 WELCOME AND LICHNEROWICZ PRIZE CEREMONY	10,00 - 11,00 MARIUS CRAINIC	10,00 - 11,00 JIANG HUA LU	10,00 - 11,00 ALBERTO IBORT	10,00 - 11,00 ÁLVARO DEL PINO
COFFEE BREAK	**COFFEE BREAK**	**COFFEE BREAK**	**COFFEE BREAK**	**COFFEE BREAK**
11,00 - 12,00 ANTON ALEKSEEV	11,30 - 12,30 YANLI SONG	11,30 - 12,30 ROBERTO RUBIO	11,30 - 12,30 HONGLEI LANG	11,30 - 12,30 CORNELIA VIZMAN
12,15- 13,15 ALEXANDER SHAPIRO	12,40 - 13,40 FRANCIS BISCHOFF	12,40 - 13,40 LUCA VITAGLIANO	12,40 - 13,40 MARCO GUALTIERI	12,40 - 13,40 YAEL KARSHON
LUNCH	**LUNCH**	**LUNCH**	**LUNCH**	**LUNCH**
15,00 - 16,00 RYSZARD NEST	15,30 - 16,30 NICOLAI RESHETIKHIN	**Free afternoon**	15,30 - 16,30 YIANNIS LOIZIDES	
16,15 - 17,15 MARÍA AMELIA SALAZAR	16,45 - 17,45 ALAN WEINSTEIN		16,45 - 17,45 DANIEL ÁLVAREZ	
			17,45 - 19,30 POSTER SESSION	
	18:30 MATEMÁTICAS EN LA RESIDENCIA		OFFICIAL DINNER	
	RECEPTION COCKTAIL			

Monday, 25 July

Venue: ICMAT CAMPUS UAM

9:00-10:00	**Bus Departure and registration at ICMAT**	
10:00–10:30	**Welcome and Lichnerowicz Prize Ceremony**	
10:30–11:00	**Coffee break**	
11:00–12:00	**Anton Alekseev** University of Geneva	Virasoro Hamiltonian spaces
12:15–13:15	**Alexander Shapiro** University of Notre Dame	Quantum decorated character stacks
	Lunch	
15:00 – 16:00	**Ryszard Nest** University of Copenhagen	Towards resolutions of singularities of Poisson structures
16:15 – 17:15	**Maria Amelia Salazar** Universidade Federal Fluminense	Relative cohomology for Lie groupoids and algebroids

Chairs: Chenchang Zhu and Cédric Oms

Tuesday, 26 July

Venue: CSIC Central Building

10:00–11:00	**Marius Crainic** Utrecht University	Constructions of codimension one symplectic foliations
11:00–11:30	**Coffee break**	
11:30–12:30	**Yanli Song** Washington University	Geometric quantization of symplectic b-manifold
12:40-13:40	**Francis Bischoff** University of Oxford	Representations of weighted homogeneous groupoids
	Lunch	
15:30–16:30	**Nicolai Reshetikhin** University of California, Berkeley and Stanford University	On superintegrability (Zoom talk)
16:45–17:45	**Alan Weinstein** University of California, Berkeley	Hamiltonian Lie algebroids over Poisson manifolds (Zoom talk
18:30	**Matemáticas en la Residencia**	
	Reception Cocktail	

Chairs: Rui Loja Fernandez and Cédric Oms

Wednesday, 27 July

Venue: CSIC Central Building

10:00–11:00	**Jiang Hua Lu** University of Hong Kong	Cluster algebras of finite type and Bott-Samelson varieties
11:00–11:30	**Coffee break**	
11:30-12:30	**Roberto Rubio** Universitat Autònoma de Barcelona	Generalized Geometry for 3-manifolds
12:40–13:40	**Luca Vitagliano** University of Salerno	Integrating Nijenhuis structures
	Lunch	
	Free afternoon	

Chair: David Iglesias

Thursday, 28 July

Venue: CSIC Central Building

10:00–11:00	**Alberto Ibort** Universidad Carlos III de Madrid	The groupoidal picture of Quantum Mechanics: the classical and semiclassical limits
11:00–11:30	**Coffee break**	
11:30-12:30	**Honglei Lang** China Agricultural University	Multiplicative multivector fields and forms on Lie groupoids
12:40-13:40	**Marco Gualtieri** University of Toronto	Generalized Kahler metrics as Lagrangian branes in double symplectic groupoids
	Lunch	
15:30–16:30	**Yiannis Loizides** Cornell University	Transversely symplectic Riemannian foliations and $[Q, R] = 0$
16:45–17:45	**Daniel Alvarez** University of Toronto	On 2-shifted lagrangian groupoids and Poisson geometry
17:45 – 19:30	**Poster Session**	
	Official dinner	

Chairs: Manuel de León, Henrique Bursztyn

Friday, 29 July

Venue: CSIC Central Building

10:00–11:00	**Alvaro del Pino** Universiteit Utrecht	Haefliger structures and wrinkling
11:00–11:30	**Coffee break**	
11:30-12:30	**Cornelia Vizman** West University of Timisoara	Coadjoint orbits relevant to ideal fluid dynamics
12:40–13:40	**Yael Karshon** Tel Aviv University and University of Toronto	Geometric quantization of Lagrangian torus fibrations
	Lunch	

Chair: Eckhard Meinrenken

Abstracts of talks (ordered according to appearance)

Virasoro Hamiltonian spaces

Anton Alekseev

Abstract: We develop a theory of Hamiltonian actions of the canonical central extension of the group of diffeomorphisms of the circle. It turns out that Virasoro Hamiltonian spaces (this is another name for such Hamiltonian actions) are in bijective correspondence with group valued Hamiltonian spaces with moment map taking values in (a certain part of) the universal cover of the group $SL(2,\mathbb{R})$. Among other things, this correspondence allows to recover the classical result of Lazutkin- Pankratova, Kirillov, Segal, Witten (and others) on classification of coadjoint orbits of the Virasoro algebra. Interesting examples of Virasoro Hamiltonian spaces arise as moduli spaces of conformally compact hyperbolic metrics on oriented surfaces with boundary.

The talk is based on a joint work in progress with Eckhard Meinrenken.

Quantum decorated character stacks

Alexander Shapiro

Abstract: I will discuss a joint work with D. Jordan, I. Le, and G. Schrader, in which we obtain categorical invariants of stratified surfaces. Our construction yields a quantization of the category of quasicoherent sheaves on the moduli stack of local systems on a surface, and uses representation theory of quantum groups and factorization homology. I will also discuss how one can recover Fock-Goncharov quantization of the moduli space of G-local systems from our construction, when G is of type A_1.

Towards resolutions of singularities of Poisson structures

Ryszard Nest

Abstract: E-symplectic manifolds include symplectic manifolds with boundary (or corners), b-symplectic manifolds, compactified cotangent bundles and regular symplectic foliations. But how general can such structures be? In this talk, I will explain how to associate an E-symplectic structure to a Poisson structure with transverse structure of semisimple type. Desingularization of these structures will also be discussed. This should allow us to address a number of open questions in Poisson Geometry from a brand-new perspective. This is joint work with Eva Miranda.

Relative cohomology for Lie groupoids and algebroids

Maria Amelia Salazar Pinzon

Abstract: In this talk I will introduce the differentiable complex of a Lie groupoid relative to a Lie subgroupoid and its analogous infinitesimal version, the basic subcomplex of the Chevalley-Eilenberg complex of a Lie algebroid. I will explain the main properties of these complexes and how they give a neat and intrinsic construction of characteristic classes for representations of Lie groupoids and Lie algebroids.

Constructions of codimension one symplectic foliations

Marius Crainic

Abstract: The existence of codimension one foliations on compact manifolds has a nice history, starting with Reeb's foliation on S^3 that marked the birth of Foliation Theory, then on general 3-manifolds, then Lawson's foliations on S^5 and then on all odd-dimensional sphere, up to the general characterization of Thurston in terms of the Euler characteristic. The analogous question for symplectic foliations is far from being understood- it is only rather recently that Lawson's foliation on S^5 was turned onto a symplectic one, by Mitsumatsu, by some rather technical arguments.

On the other hand, the confoliations of Eliashberg-Thurston revealed very close relationship (via actual deformations) between foliations and contact structure. That theory is well developed only in dimension 3 and that is due, I believe (at least in part), to not noticing that when moving to higher dimensions one should be looking not only at foliations, but at symplectic ones. As a side remark: also Mitsumatsu's construction exploits the geometry of contact forms and adapted open book decompositons.

In this talk I will provide an overview of various existing constructions of symplectic foliations, then I will present a new method- via stable generalized complex geometry. In the case of Lawson's foliation on S^5, the new method provides a much more direct and simpler (we believe) way of turning it into a symplectic foliation. This is based on joint work with various people in Utrecht, among which my colleague Gil Cavalcanti and the former PhD students Lauran Toussaint and Aldo Witte.

Geometric quantization of symplectic b-manifold

Yanli Song

Abstract: In this talk, I will discuss one method of geometric quantization for compact b-symplectic manifolds in terms of the index of Dirac operators. Under an orientation hypothesis, our method also applies to manifold with simple normal crossing singularities. I will also talk about the "quantization commutes with reduction" principle in this setting. This is a joint work with Braverman, Loizides, Lin and Sjamaar.

Representations of weighted homogeneous groupoids

Francis Bischoff

Abstract: In this talk I will present results on the representation theory of weighted homogeneous groupoids. This is a family of holomorphic Lie groupoids that are closely related to torus action groupoids, and in many examples, their representations correspond to flat connections with logarithmic singularities along Saito free divisors. As an application of the theory, I will present resonant normal form theorems for flat log connections for several examples of free divisors, as well as results related to the local Riemann-Hilbert problem

On superintegrability

Nicolai Reshetikhin

Abstract: The talk will start with a short overview of superintegrable systems in Hamiltonian mechanics and in quantum mechanics. Then we will focus on examples related to moduli spaces of flat connections and their quantization. It will be concluded with a discussion of open problems.

Hamiltonian Lie algebroids over Poisson manifolds

Alan Weinstein

Abstract: Christian Blohmann and I, along with Stefano Ronchi, are continuing our study of hamiltonian Lie algebroids, extending now from symplectic manifolds to Poisson manifolds rather than presymplectic manifolds. Many of the basic ideas remain the same, but there are many new examples, including the cotangent Lie algebroid of a Poisson manifold.

The Poisson case is also the most appropriate setting to test (finally!) some ideas proposed by Alejandro Cabrera after a talk which I gave in Banff in 2017. These concern the possibility of interpreting the compatibility of a Lie algebroid structure, a connection, and a Poisson structure in terms of a naturally associated pair of Poisson structures on the dual of the Lie algebroid.

Cluster algebras of finite type and Bott-Samelson varieties

Jiang Hua Lu

Abstract: For a complex simple Lie group G and a Coxeter element c in the Weyl group of G, the reduced double Bruhat cell $L^{c,c^{-1}}$ in G has a cluster structure of finite type. We use an embedding of $L^{c,c^{-1}}$ into a Bott-Samelson variety to construct all the cluster tori in $L^{c,c^{-1}}$, and we explain the Poisson geometry in the construction.

Generalized Geometry for 3-manifolds

Roberto Rubio

Abstract: Generalized complex geometry offers a unifying approach to complex and symplectic structures. Moreover, there are generalized complex manifolds that are neither complex nor symplectic, but any of them must be almost complex and hence even-dimensional. In this talk I will discuss how Bn- generalized geometry, the simplest variation of generalized geometry, provides a similar framework for manifolds of any dimension and I will especially focus on the case of 3-manifolds, which is recent joint work with J. Porti.

Integrating Nijenhuis Structures

Luca Vitagliano

Abstract: A Nijenhuis operator on a manifold is a (1,1) tensor fulfilling an appropriate integrability condition. Nijenhuis operators appear naturally in complex geometry, integrable systems, projective geometry, etc. Recently, they draw the attention of researchers in their own. A Nijenhuis operator N defines a Lie algebroid which knows everything about N. We say that a Lie groupoid G integrates N if G integrates the Lie algebroid of N. In this case, G can be seen as a global counterpart of N, similarly as symplectic groupoids are global counterparts of Poisson structures. In this talk we characterize Lie groupoids integrating a Nijenhuis operator. Our criterion involves a cochain complex whose 0-cocycles are multiplicative (1,1) tensors. Time permitting, we will illustrate our main result with various simple examples including: integrable almost tangent structures, integrable projections, 2-step nilpotent Lie algebras (joint work with F. Pugliese and G. Sparano).

The groupoidal picture of Quantum Mechanics: the classical and semiclassical limits

Alberto Ibort

Abstract: Schwinger's picture of Quantum Mechanics fits perfectly in a categorical/groupoidal description. Building upon this idea the main notions of quantum mechanics will be succinctly reviewed and a novel interpretation of the quantum-to-classical transition will be discussed. In particular, it will be shown how Schwinger's quantum dynamical principle leads to Lagrangian dynamics on Lie algebroids and its dual Hamiltonian formulation.

Multiplicative multivector fields and forms on Lie groupoids

Honglei Lang

Abstract: We first present two formulas in which multiplicative multivector fields (abbr. to MVs) and multiplicative forms (abbr. to MFs) on Lie groupoids are decomposed, respectively, into the sum of two parts, and we call the data obtained characteristic pairs. Then for a quasi-Poisson groupoid, we show that the space of MFs, together with ordinary forms on the base manifold, admits a graded weak Lie 2-algebra. The quasi-Poisson structure further builds a morphism from the said graded weak Lie 2-algebra to the graded Lie 2-algebra of MVs studied by Bonechi-Ciccoli-Laurent-Gengoux-Xu. This is joint work with Zhuo Chen and Zhangju Liu.

Generalized Kahler metrics as Lagrangian branes in double symplectic groupoids

Marco Gualtieri

Abstract: Generalized Kahler manifolds, which arise as natural target spaces for supersymmetric 2-dimensional sigma models, provide a rich family of examples to which many of the tools of Poisson geometry may be applied. After reviewing their basic structure, I will explain a new appearance of double groupoids and double Morita equivalences in the story. This is joint work with Yucong Jiang.

Transversely symplectic Riemannian foliations and $[Q, R] = 0$

Yiannis Loizides

Abstract: I will describe a quantization commutes with reduction ($[Q, R] = 0$) theorem for Riemannian foliations equipped with a transverse symplectic structure. This is joint work with Yi Lin, Reyer Sjamaar, and Yanli Song.

On 2-shifted lagrangian groupoids and Poisson geometry

Daniel Àlvarez

Abstract: In this talk we will explain how the language of shifted symplectic geometry sheds some light on several interesting examples in Poisson geometry such as (1) the integrations of Poisson homogeneous spaces, (2) the configuration groupoids of flags and (3) the global counterparts of quasi-Poisson spaces. Joint work with H. Bursztyn and M. Cueca.

Haefliger structures and wrinkling

Alvaro Del Pino

Abstract: Foliations in open manifolds were classified by Haefliger in "Feuilletages sur les variétés ouvertes". Using Gromov's method of flexible sheaves, he proved that the space of foliations (up to concordance) is weakly equivalent to the space of Gamma structures (i.e. cocycles valued in the étale Lie groupoid of diffeomorphism germs of Euclidean space), which may be thought of as foliations with singularities.

Gamma structures also appear naturally in the study of geometric structures other than foliations. From the perspective of Geometry of PDEs, this has been explored in the works of M. Crainic, M.A. Salazar, O. Yulidevich, F. Cattafi, and L. Accornero, among others. In a more homotopical direction, Haefliger himself observed that Gamma structures can be used to state classification results for symplectic and complex structures.

This latter perspective was recently pursued by Laudenbach and Meigniez. In "Haefliger structures and symplectic/contact structures", they present the construction of geometric structures as a two-step process. First, one attempts to construct a Gamma structure starting from some given formal data. Then, one tries to produce a genuine geometric structure starting from a Gamma structure. Their main result says that the first step can always be carried out in the contact/symplectic case.

In this talk I will explain how the wrinkling approach of Eliashberg-Mishachev may be used to produce Gamma-structures for any family of geometric structures given by an open and Diff-invariant partial differential relation. I will put particular emphasis in explaining how h- principle ideas combine naturally

with the Lie groupoid formalism needed to set up Gamma structures.

If time allows, I will explain the role played by this result in the classification problem for contact structures and conformal symplectic foliations.

This is on-going work with A. Fokma and L. Toussaint.

Coadjoint orbits relevant to ideal fluid dynamics

Cornelia Vizman

Abstract: We explore coadjoint orbits for the group of volume preserving diffeomorphisms that are modeled on nonlinear Grassmannians of ubmanifolds of codimension one. We focus on vortex loops in 2D (collections of point vortices) and on vortex sheets in 3D (collections of vortex filaments). We also determine a Feynman-Onsager condition for their prequantizability.

Geometric quantization of Lagrangian torus fibrations

Yael Karshon

Abstract: By the Arnold-Liouville theorem, a regular completely integrable system on a compact symplectic manifold M is Lagrangian torus fibration over an integral affine manifold B. When M is pre-quantized, we give a simple proof that its Riemann-Roch number coincides with its number of Bohr-Sommerfeld leaves. This can be viewed as an instance of the mysterious "independence of polarization" phenomenon of geometric quantization.

Abstract of posters

Equivariant cohomology for 2-group actions

Luis Alejandro Barbosa-Torres

We consider a strict action of a Lie 2-group on a Lie groupoid and we look for a notion of equivariant cohomology. Consequently, we check this cohomology is Morita invariant, giving us the relation with the category of differentiable stacks.

A Hamiltonian Floer complex for b-symplectic manifolds

Joaquim Bruguès

Since the beginnings of the development of Floer theory there has been an interest to adapt this construction to non-compact manifolds. In our project, we aim at building one such complex for the open part of b-symplectic manifolds, using a cosymplectic foliation in a tubular neighbourhood near the singular hypersurface. This is a joint project with Eva Miranda and Cédric Oms.

Existence and classification of b-contact structrues

Robert Cardona

One of the simplest class of non-transitive Jacobi structures are b-Jacobi structures, which are singular along a hypersurface. These were first studied by Miranda-Oms, using Melrose's b-calculus to reformulate them as b-contact structures. Using

this approach and techniques from contact topology, we state general existence results for b-contact structures on any oriented three dimensional b-manifold (M, Z), i.e. prescribing the singular hypersurface Z. We completely classify b-contact structures, up to isotopy, in some concrete b-manifolds. This is joint work with Cédric Oms.

Extending the Large Color R-matrix to the three-fold symmetric Lie algebra $\mathfrak{so}(8)$

Alberto Ricardo Cavallar Oriol

The Gukov-Manolescu series, denoted by $FK(x, q)$, is a conjectural invariant of knot complements that, in a sense, analytically continues the colored Jones polynomials. In a recent paper, Sunghyuk Park uses the large color R-matrix of $\mathfrak{sl}(2)$ to study FK for some simple links, giving a definition of FK for positive braid knots and computing FK for various knots and links. This procedure has been extended by Angus Gruen to all other Lie algebras $\mathfrak{sl}(n+1)$ beyond $\mathfrak{sl}(2)$. In this work, we turn to the family $\mathfrak{so}(2n)$ of complex semisimple Lie algebras in Cartan's classification, mainly focusing on the $\mathfrak{so}(8)$ case attracted by the three-fold symmetry in its Dynkin diagram $D4$.

Geometric Integrators in Poisson Geometry

Oscar Cosserat

We use local symplectic Lie groupoids to construct Poisson integrators for generic Poisson structures. More precisely, recursively obtained solutions of a Hamilton-Jacobi-like equation are

interpreted as Lagrangian bisections in a neighborhood of the unit manifold, that, in turn, give Poisson integrators.

Homotopy momentum sections and sigma models

Noriaki Ikeda

We introduce a notion of a homotopy momentum section on a Lie algebroid over a pre-multisymplectic manifold. A homotopy momentum section is a generalization of the momentum map with a Lie group action and the momentum section on a pre-symplectic manifold, and is also a generalization of the homotopy momentum map on a multisymplectic manifold. A gauged nonlinear sigma model with Wess-Zumino term with Lie algebroid gauging has the homotopy momentum section structure.

Log Floer cohomology for log symplectic surfaces

Charlotte Kirchhoff-Lukat

I define a notion of Lagrangian intersection Floer cohomology for (orientable) log symplectic surfaces and describe the interaction between different symplectic components in this theory. I show that for a single Lagrangian, this cohomology is isomorphic to log de Rham cohomology. I will discuss what this low-dimensional toy model can teach us about Floer cohomology for stable generalized complex structures.

Symmetries of singular foliations through universal Lie infinity algebroids

Ruben Louis

We show that there is an equivalence of categories between Lie-Rinehart algebras over a commutative algebra O and homotopy equivalence classes of negatively graded Lie infinity-algebroids over their resolutions. This extends to a purely algebraic setting the construction of the universal Q-manifold of a locally real analytic singular foliation. In particular, it makes sense for the universal Lie infinity-algebroid of every singular foliation, without any additional assumption, and for Androulidakis-Zambon singular Lie algebroids. Also, to any ideal $I \subset O$ preserved by the anchor map of a Lie-Rinehart algebra A, we associate a homotopy equivalence class of negatively graded Lie infinity-algebroids over complexes computing $\operatorname{Tor} O(A, O/I)$.

Poisson Structures on Sets of Maurer-Cartan Elements

Thomas Machon

Given a differential graded Lie algebra (dgla) L satisfying certain conditions, we construct Poisson structures on the gauge orbits of its set of Maurer-Cartan (MC) elements, termed Maurer-Cartan-Poisson (MCP) structures. They associate a compatible Batalin-Vilkovisky algebra to each MC element of L. An MCP structure is shown to exist for a number of dglas associated to commutative Frobenius algebras, deformations of Poisson and symplectic structures, as well as the Chevally-Eilenberg complex. MCP structures yield a notion of hamiltonian flow of MC

elements, and also define Lie algebroids on gauge orbits, whose isotropy algebras give invariants of MC elements. As an example, this gives a finite-dimensional two-step nilpotent graded Lie algebra associated to any closed symplectic manifold.

Shifted Contact Structures on Differentiable Stacks

Antonio Maglio

♯ A contact structure on a manifold is equivalent to a line bundle L equipped with a non-degenerate, L-valued 2-cocycle W in the gauge algebroid DL (also called a symplectic Atiyah form).

In analogy with the symplectic case, we define a $+1$ shifted contact structure on a Lie groupoid G as a line bundle groupoid $L \to G$ together with an Atiyah 2-form W on L such that

- W is multiplicative
- dW is cohomologically trivial wrt the simplicial cohomology
- the flat map of W is a Morita map of VB groupoids

Our main result is a stacky version of ♯. As a side result, we prove that the kernel of a multiplicative 1-form with values in a trivial core VB groupoid $E \to G$ (may not exist as a smooth multiplicative distribution on G, but it) always exists as a stacky distribution on the differentiable stack presented by G. The proof involves a VB groupoid version of the homotopy kernel construction in Homological Algebra. This is joint work with Alfonso Giuseppe Tortorella and Luca Vitagliano.

The hamiltonian formulation of classical gauge theories on E-manifolds

Pablo Nicolás Martínez

We extend the formalism of vector bundles, principal bundles and principal connections to E-manifolds (as introduced in Nest and Tsygan), which can be used to describe singularities in the configuration space of a classical particle. Manifolds with boundary or corners are configuration spaces naturally described in terms of E-manifolds extending that of b-manifolds (see Guillemin, Miranda, and Pires or Gualtieri and Li). Following Weinstein, we show the existence of a universal model for the phase space of a particle interacting with a gauge field; in this new setting, Wong's equations become hamiltonian. Also following Montgomery, we see that the universal E-symplectic spaces of Weinstein for Yang-Mills theories are symplectic leaves of a bigger universal Poisson space. This is a joint work with Pau Mir and Eva Miranda.

Bohr-Sommerfeld quantization of b-symplectic toric manifolds

Pau Mir García

We define the Bohr-Sommerfeld quantization via T-modules for a b-symplectic toric manifold and show that it coincides with the formal geometric quantization of Guillemin, Miranda and Weitsman. In particular, we prove that its dimension is given by a signed count of the integral points in the moment polytope of the toric action on the manifold.

Quasifolds as groupoids and diffeological spaces

David Miyamoto

A quasifold is a space that is locally modeled by quotients of $\mathbb{R}^n$ by countable group actions. These arise in Elisa Prato's generalization of the Delzant theorem to irrational polytopes, and include irrational tori, orbifolds, and certain leaf spaces of foliations. We approach quasifolds in two ways: by viewing them as diffeological spaces, we form the category of diffeological quasifolds, and by viewing them as Lie groupoids, we form the category of quasifold groupoids. We show that an effective quasifold groupoid is determined by its diffeological orbit space. This is joint work with Yael Karshon.

Palatini formalism in the framework of Courant algebroids

Filip Moučka

Palatini approach to the Einstein-Hilbert action is a way how to justify the use of the Levi- Civita affine connection for general relativity. It has turned out that kind of similar thing can be done also for the Levi-Civita Courant algebroid connections. In the particular case of exact Courant algebroids, this procedure leads precisely to the Courant algebroid formulation of the supergravity (SUGRA) equations.

Courant-Nijenhuis algebroids

Claire Netto

Inspired by Poisson-Nijenhuis structures, we define Courant-Nijenhuis (CN) algebroids using the formalism of generalized derivations on vector bundles. From this approach, we introduce a notion of Dirac-Nijenhuis (DN) structures which recovers the definition of Poisson-Nijenhuis structures when the Courant algebroid is the Pontryagin bundle $TM + T^*M$. We explore the examplos of holomorphic Courant algebroids and CN structures on $TM + T^*M$. Moreover, we revisit the theory of Manin triples for Lie algebroids, and establish a relation between Lie-Nijenhuis bialgebroids, CN algebroids and DN structures. Joint work with Henrique Bursztyn and Thiago Drummond.

Higher Cotangent groupoids

Stefano Ronchi

It is well known that there exists a cotangent object for any Lie groupoid, which corresponds to the coadjoint representation up to homotopy, and that it is a symplectic groupoid. Our work aims to answer the question of how an object of this kind can be constructed for Lie n-groupoids, and more generally, what the dual objects of VB-n-groupoids are. This is a summary of several projects joint with Miquel Cueca, Madeleine Jotz Lean and Chenchang Zhu.

Stability results in Poisson geometry and higher Lie theory

Karandeep Singh

When studying deformations of a structure, a natural question one can ask is when some property is preserved when look-

ing at a neighborhood of a given structure. For example, let Q be a structure which induces a singular foliation on the manifold M, such as a Lie ($n-$)algebroid, Courant algebroid or Poisson structure. Assume that $p \in M$ is a singular point of the structure. Then one can ask is when any structure Q' near Q has a singular point near p.

We show that these problems are an instance of a more general question: given a differential graded Lie algebra $(\mathfrak{g}, \partial, [-,-])$, a differential graded Lie subalgebra $(\mathfrak{h}, \partial, [-,-])$, and a Maurer-Cartan element $Q \in \mathfrak{h}^1$, when is it the case that for every $Q' \in \mathfrak{g}^1$, there exists $X \in \mathfrak{g}^0$, such that the gauge transformation of Q' by X lies in $\mathfrak{h}^1$?

When the codimension of $\mathfrak{h}$ in $\mathfrak{g}$ is finite, a sufficient condition for a positive answer is that $H^1(\mathfrak{g}/\mathfrak{h}, \partial + [Q,-]) = 0$. In the case of Lie algebroids and Poisson structures, this recovers the Crainic-Fernandes stability criteria for zero-dimensional leaves.

Lie algebroid descriptions of T-duality

Juan José Villamarín

T-duality establishes an equivalence between two field theories on different space-time manifolds, it has been well understood in the context of principal T-bundles [BHM05]. In this context, the space-time is modeled by a principal T-bundle E over M and E is endowed with the so-called H-flux, a closed T-invariant 3-form H. Indeed, two different space-times (E, H) (E', H') are T-dual if $H - H'$ is exact in the correpondence space $Ex_M E'$. In this work we give an interpretation of T-duality in the context of Courant and Lie algebroids and exlpore some of its consequences. In particular, we show that T-duality can

be understood as an isomorphism of Courant algebroids [CG10] and, furthermore, we show that T-duality can be seen as a duality of Atiyah algebroids $A(E)$ and $A(E')$, i.e. they define a Lie bi-algebroid [Mac05].

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Generalizing b^k-geometry

Aldo Witte

The b-tangent bundle is the Lie algebroid consisting of vector fields tangent to a given embedded hypersurface Z. Although it is locally clear what is meant with vector fields tangent up to order k at Z, giving a global definition is surprisingly subtle. One approach is given by Scott where the auxiliary data of a jet of defining function is used. We generalized the notion of Scott, by considering the data of a jet of foliation F around Z and considering vector fields tangent to F at Z up to order k. Using an explicit description of the holonomy of these Lie algebroids, we show that any Lie algebroid which is locally given by vector

fields tangent up to order k at Z, is of this form. Along the way, we will show that variations of this construction allow one to recover many other Lie algebroids and singular distributions. Joint work with Alvaro del Pino.

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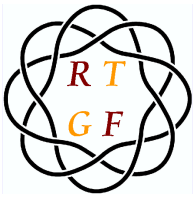

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